CN - CheatSheet

Compiled BY : Manthan A. D. [in]

CHAPTER 1 - INTRODUCTION TO COMPUTER NETWORKS

1. Network fundamentals

- **Computer network:** Interconnected devices that exchange data and share resources.
- **Node:** A device connected to a network.
- **Host:** A network-connected device that sends or receives data.
- **Protocol:** Rules that govern communication.
- **NIC (Network Interface Card):** Hardware/virtual interface that connects a device to a network.
- **Client:** Requests a service.
- **Server:** Provides a service.
- **Peer-to-peer (P2P):** Devices can act as both clients and providers.
- **Network topology:** Physical or logical arrangement of network devices.
- **Network architecture:** Organization of network components, layers and protocols.

2. Types of networks

Type	Full form	Typical coverage / example
PAN	Personal Area Network	Personal devices; Bluetooth
LAN	Local Area Network	Room, home, office, building
WLAN	Wireless LAN	Wi-Fi network
CAN	Campus Area Network	College or corporate campus
MAN	Metropolitan Area Network	City or metropolitan area
WAN	Wide Area Network	Large geographic areas; countries
GAN	Global Area Network	Global connectivity

- **LAN:** Usually privately managed and high-speed over a limited area.
- **MAN:** Connects networks across a city.
- **WAN:** Connects networks over large distances.
- **Internet:** Global interconnection of networks, not one single WAN.

3. Network models

OSI model

Layer (top to bottom)	Main function	PDU
7. Application	Network services to applications	Data
6. Presentation	Translation, encryption, compression	Data
5. Session	Session management	Data
4. Transport	Process-to-process delivery, reliability/flow control where supported	Segment / Datagram
3. Network	Logical addressing and routing	Packet
2. Data Link	Framing, MAC addressing, link error detection	Frame
1. Physical	Transmits bits as signals	Bits

Mnemonic, top to bottom: **All People Seem To Need Data Processing.**

TCP/IP model

1. **Application:** OSI Application + Presentation + Session.

2. **Transport:** TCP/UDP and process delivery.
3. **Internet:** IP addressing and routing.
4. **Network Access:** OSI Data Link + Physical.

- **OSI:** Seven-layer reference model.
- **TCP/IP:** Four-layer practical Internet protocol suite.

Encapsulation

Sender: **Data** → **Segment/Datagram** → **Packet** → **Frame** → **Bits**.
Receiver: Reverse process, called **decapsulation**.

4. Network topologies

Topology	Main idea	Key point
Star	All devices connect to a central switch/hub	Easy troubleshooting; central-device failure matters
Bus	Devices share a backbone	Simple; backbone fault affects network
Ring	Devices form a loop	Data follows ring arrangement
Mesh	Devices have multiple interconnections	Reliable but expensive
Tree	Hierarchical arrangement	Scalable, hierarchical management

5. Transmission media

- **Guided media:** Signals travel through a physical medium.
  - Twisted pair: common in Ethernet LANs.
  - Coaxial cable: shielded copper cable.
  - Optical fibre: light-based, high capacity, immune to electromagnetic interference.
- **Unguided media:** Wireless transmission through space.
  - Radio waves: Wi-Fi, Bluetooth, cellular.

- Microwaves: point-to-point wireless links.
- Infrared and satellite links: specialized wireless uses.

6. Transmission modes

- **Simplex:** One-way only.
- **Half-duplex:** Both directions, one at a time. Example: walkie-talkie.
- **Full-duplex:** Both directions simultaneously. Example: telephone call.

7. Networking devices and addresses

Device / address	Function
Repeater	Regenerates signals
Hub	Sends incoming bits to all ports
Bridge	Connects LAN segments using MAC information
Switch	Forwards frames using MAC addresses
Router	Forwards packets using IP addresses
Gateway	Connects dissimilar networks/protocols
Modem	Modulates/demodulates signals for a communication link
Access Point	Connects wireless devices to a LAN
MAC address	Link-layer interface address
IP address	Logical network-layer address
Port number	Identifies a transport-layer service endpoint

Remember: **Switch → MAC; Router → IP.**

8. Switching and service types

- **Circuit switching:** Dedicated circuit/path established for a session.
- **Packet switching:** Packets share network resources.
- **Store-and-forward:** Receive a complete packet/frame before forwarding it.
- **Datagram network:** Each packet may be routed independently.

- **Virtual circuit:** A logical route is established before data transfer.
- **Connection-oriented service:** Connection/state established before data transfer; TCP is an example.
- **Connectionless service:** No transport connection setup; UDP is an example.

9. Performance terms and formulas

- **Bandwidth:** Channel capacity, often expressed in bits per second.
- **Throughput:** Actual successful data transfer rate.
- **Latency:** Time taken for data to travel.
- **Jitter:** Variation in packet delay.
- **Packet loss:** Packets that fail to reach the destination.
- **Bit rate:** Bits per second (bps).
- **Baud rate:** Symbols per second (baud).
- **Bit rate = Baud rate × bits per symbol.**

Four delay types

- Transmission delay: **d_trans = L / R**
- Propagation delay: **d_prop = D / S**
- Processing delay: Time to inspect/process a packet.
- Queuing delay: Time spent waiting in a queue.
- Total nodal delay = processing + queuing + transmission + propagation.
- L = packet size in bits; R = link rate in bps; D = distance; S = propagation speed.

CHAPTER 2 — DATA LINK AND NETWORK LAYER

1. Data Link Layer essentials

- **Framing:** Divides a bit stream into identifiable frames.
- **Physical addressing:** MAC addressing for local-link delivery.
- **Flow control:** Prevents sender from overwhelming receiver.
- **Error control:** Detects errors and supports recovery.
- **MAC (Media Access Control):** Coordinates access to a shared medium.
- **LLC (Logical Link Control):** Logical-link functions in IEEE 802 architecture.

2. Error detection and correction

- **Parity check:** Adds a bit to make the number of 1s even or odd.
  - Even parity: total 1s are even.
  - Odd parity: total 1s are odd.
  - Single parity detects odd numbers of bit errors but can miss even numbers.
  - Two-dimensional parity can locate/correct certain single-bit errors.
- **Checksum:** Calculated value used to help detect corruption.
- **CRC (Cyclic Redundancy Check):**
  1. Append r zeros to data, where r = generator-polynomial degree.
  2. Divide using modulo-2 division (XOR, no carries/borrows).
  3. Remainder is the CRC.
  4. Replace appended zeros with remainder to form codeword.
  5. Receiver divides by same generator; zero remainder means no error was detected.
- **Hamming distance:** Number of bit positions in which two bit strings differ.
  - Detect up to d_min − 1 errors.
  - Correct up to floor((d_min − 1)/2) errors.
- **Bit stuffing:** Insert a 0 after five consecutive 1s in flag-based protocols such as HDLC.

- **Byte stuffing:** Insert escape bytes when reserved bytes occur in data.

3. MAC and channel-access protocols

- **ALOHA:** Transmit without first sensing the channel.
- **Pure ALOHA:** Transmit anytime; maximum theoretical efficiency ≈ 18.4%.
- **Slotted ALOHA:** Transmit at slot boundaries; maximum theoretical efficiency ≈ 36.8%.
- **CSMA:** Sense the channel before transmitting.
- **1-persistent CSMA:** Transmit immediately when the channel becomes idle.
- **Non-persistent CSMA:** If busy, wait a random time and sense again.
- **p-persistent CSMA:** On an idle channel, transmit with probability p.
- **CSMA/CD:** Detects collisions; associated with traditional shared, half-duplex Ethernet.
- **CSMA/CA:** Attempts to avoid collisions; used in Wi-Fi.
- **Polling:** A controller invites devices to transmit.
- **Token passing:** A device transmits when it holds the token.

4. Ethernet, LAN and VLAN

- **Ethernet:** Common wired LAN technology; IEEE 802.3.
- Ethernet frame fields: **Destination MAC** → **Source MAC** → **Type/Length** → **Data + Padding** → **FCS**.
- **FCS:** Frame Check Sequence used for error detection.
- **VLAN:** Logically divides a switched network into separate broadcast domains.
- **Broadcast domain:** Set of devices that receive a link-layer broadcast.

5. IP addressing

- **IPv4:** 32-bit address, dotted-decimal notation.
- **IPv6:** 128-bit address, hexadecimal colon-separated notation.

- **Subnet mask:** Separates network bits from host bits.
- **CIDR:** Classless addressing using prefix notation, e.g. /24.
- **VLSM:** Uses different subnet sizes according to host requirements.
- **Default gateway:** Router used to reach destinations outside the local subnet.
- **Loopback:** IPv4 127.0.0.1; IPv6 ::1.
- **APIPA/link-local IPv4:** 169.254.0.0/16.

Traditional IPv4 classes

Class	First octet	Default mask / purpose
A	1–126*	/8
B	128–191	/16
C	192–223	/24
D	224–239	Multicast
E	240–255	Experimental/reserved

127.x.x.x is loopback. Modern networks primarily use CIDR, not classful allocation.

Private IPv4 ranges

- 10.0.0.0/8
- 172.16.0.0/12
- 192.168.0.0/16

Subnetting formulas

- IPv4 address length = 32 bits.
- Total addresses = **2^(32 – p)**.
- Host bits = **32 – p**.
- Conventional usable hosts = **2^h – 2**, where h = host bits.
- Block size in the relevant octet = **256 – subnet-mask value**.
- /24 = 255.255.255.0.

- /26 = 255.255.255.192: 64 total, usually 62 usable.
- /27 = 255.255.255.224: 32 total, usually 30 usable.

6. IPv4 header and related protocols

IPv4 header fields to recognize in a diagram:

- Version, IHL, ToS/DS field, Total Length.
- Identification, Flags, Fragment Offset.
- TTL, Protocol, Header Checksum.
- Source IP, Destination IP.
- Options and Padding (optional).
- **TTL:** Limits packet lifetime.
- **Protocol field:** Identifies upper-layer payload, e.g. TCP or UDP.
- **Fragmentation:** Splits an IPv4 packet when required by link/path constraints.
- **ARP:** Resolves IPv4 address to a local-link MAC address.
- **ICMP:** Error reporting and diagnostics, e.g. ping.
- **DHCP:** Automatically assigns IP configuration.
- **NAT:** Translates addresses between address spaces, often allowing multiple devices to share a public IP.
- **Unicast:** One sender to one receiver.
- **Multicast:** One sender to a selected group.
- **Broadcast:** One sender to all devices in the broadcast domain.
- **Physical address:** MAC, Data Link layer.
- **Logical address:** IP, Network layer.
- **Port address:** Process/service endpoint, Transport layer.

IPv4 vs IPv6

IPv4	IPv6
32-bit address	128-bit address
Dotted decimal	Hexadecimal, colon-separated
Header checksum present	No base-header checksum
Broadcast supported	No broadcast; multicast is used

7. Routing

- **Routing:** Selecting a path for packets.
- **Static routing:** Configured manually.
- **Dynamic routing:** Routers learn/update routes automatically.
- **Distance Vector Routing (DVR):** Routers exchange distance/next-hop information with neighbors.
- **Link State Routing (LSR):** Routers distribute link-state information and compute shortest paths.
- **Dijkstra:** Shortest-path algorithm used by link-state routing.
- **Bellman-Ford principle:** Underlies distance-vector route calculation.

Distance Vector formula

$D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$

- $D_x(y)$: best known cost from router x to destination y.
- $c(x,v)$: cost from x to neighbor v.
- Routers update tables until convergence.

DVR steps and count-to-infinity

1. Initialize known direct-link costs.
2. Exchange distance vectors with neighbors.
3. Recalculate routes using neighbor costs.
4. Update next hops and repeat until stable.

- **Count-to-infinity:** After a route failure, routers may keep increasing a bad route's metric.
- Solutions include split horizon, poison reverse and route-update techniques.

Link State Routing steps

1. Discover neighbors.
2. Measure link costs.
3. Flood link-state information.
4. Build topology map.
5. Run Dijkstra's algorithm.
6. Construct routing table.

Important routing protocols

Protocol	Type / key fact
RIP	Distance-vector; hop-count metric
OSPF	Link-state; cost metric
BGP	Path-vector; inter-domain routing

CHAPTER 3 — TRANSPORT AND APPLICATION LAYER

1. Transport layer basics

- **Segment:** TCP transport-layer PDU.
- **Datagram:** UDP transport-layer PDU.
- **Multiplexing:** Combines data from applications for transport.
- **Demultiplexing:** Delivers incoming data to the correct socket/application.
- **Port number:** 16-bit identifier used by transport protocols.

- **Socket:** Communication endpoint, commonly associated with IP address and port.
- **Process-to-process delivery:** Transport layer uses port numbers to deliver data to applications.

Port ranges

- 0–1023: Well-known ports.
- 1024–49151: Registered ports.
- 49152–65535: Dynamic/private ports.

2. TCP vs UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable, ordered byte stream	No delivery/order guarantee
Flow and congestion control	No built-in transport flow/congestion control
Higher overhead	Lower overhead
Web, email, file transfer	DNS, real-time media, gaming

3. TCP header and flags

Header fields: Source Port, Destination Port, Sequence Number, Acknowledgment Number, Header Length, Flags, Window Size, Checksum, Urgent Pointer, Options and Padding.

Flag	Meaning
SYN	Establish connection
ACK	Acknowledgment field is valid
FIN	Graceful close
RST	Reset connection
PSH	Request prompt delivery to application
URG	Urgent pointer is significant

- **Sequence number:** Identifies byte positions in the stream.
- **Acknowledgment number:** Indicates next expected byte.
- **Checksum:** Helps detect corruption.
- **Receive window:** Advertises receiver capacity for flow control.
- **Retransmission:** Resends data presumed lost.
- **Cumulative ACK:** Acknowledges contiguous received data.

4. TCP connection management

Three-way handshake

1. Client → Server: SYN.
2. Server → Client: SYN + ACK.
3. Client → Server: ACK.

- Termination commonly uses FIN/ACK exchanges in both directions.
- RST can reset a connection.

5. Flow, error and congestion control

- **Flow control:** Prevents overwhelming the receiver.
- **Error control:** Detects/recover from loss or corruption.
- **Congestion control:** Prevents overwhelming the network.
- **Slow Start:** Congestion window grows exponentially, approximately doubling each RTT.
- **Congestion Avoidance:** Window grows approximately linearly.
- **Fast Retransmit:** Typically retransmits after 3 duplicate ACKs.
- **Fast Recovery:** In applicable TCP algorithms, reduces sending rate without restarting from the initial window.

6. Flow-control protocols

- **Stop-and-Wait:** Send one frame, then wait for ACK.
- **Sliding Window:** Multiple frames can be outstanding before ACKs arrive.
- **Go-Back-N (GBN):** Cumulative ACKs; retransmits missing frame and subsequent outstanding frames; receiver generally discards out-of-order frames.
- **Selective Repeat (SR):** Individually acknowledges frames, buffers valid out-of-order frames, retransmits only missing frames.
- For m-bit sequence numbers:
  - GBN maximum sender window = 2^m − 1.
  - SR maximum window = 2^(m−1).
- **Piggybacking:** Attach an ACK to outgoing data instead of sending a separate ACK immediately.
- **RDT (Reliable Data Transfer):** Protocol principles for handling corruption, loss, acknowledgments and retransmission.

7. DNS

- **DNS:** Converts domain names to IP addresses.
- Hierarchy: Root → TLD → Authoritative domain server.
- **Recursive query:** Resolver obtains final answer on behalf of client.
- **Iterative query:** Server returns answer or referral to another server.
- **Caching:** Stores results to reduce repeated lookups.
- **FQDN:** Fully qualified domain name.
- **PQDN:** Partially qualified domain name.

8. HTTP and HTTPS

- **HTTP:** Application-layer request-response protocol.
- **Stateless:** Protocol does not inherently retain request state between separate requests.

- **Persistent connection:** Reuses a connection for multiple exchanges.
- **Non-persistent connection:** A separate connection is used for each exchange.
- **HTTPS:** HTTP secured using TLS.

Request structure: **Request line** → **Headers** → **Blank line** → **Optional body**.
Response structure: **Status line** → **Headers** → **Blank line** → **Optional body**.

Common HTTP status codes

- 200: OK.
- 301/302: Redirection.
- 400: Bad Request.
- 401: Unauthenticated.
- 403: Forbidden.
- 404: Not Found.
- 500: Internal Server Error.
- 503: Service Unavailable.

9. Email and application protocols

- **SMTP:** Sends outgoing email and transfers mail between servers.
- **POP3:** Retrieves email, often downloading for local storage.
- **IMAP:** Manages/synchronizes mail on the server.
- **MIME:** Supports attachments and multimedia/non-ASCII email content.
- **FTP:** File transfer.
- **SSH:** Secure remote login.
- **Telnet:** Unencrypted remote terminal access.
- **REST API:** Commonly uses HTTP methods such as GET, POST, PUT and DELETE to work with resources.

Well-known ports

Protocol	Port
FTP control	21
SSH	22
Telnet	23
SMTP	25
DNS	53
HTTP	80
POP3	110
HTTPS	443

10. Socket programming

TCP server

socket() → bind() → listen() → accept() → send()/recv() → close()

TCP client

socket() → connect() → send()/recv() → close()

- **socket():** Creates communication endpoint.
- **bind():** Assigns local IP/port.
- **listen():** Marks TCP socket ready for incoming connections.
- **accept():** Accepts incoming connection.
- **connect():** Initiates connection to server.
- **send()/recv():** Send/receive data.

CHAPTER 4 — WIRELESS, IoT AND CLOUD NETWORKING

New-syllabus chapter. The supplied PYQ analysis reports no previous-paper coverage for this chapter; study the official syllabus topics anyway.

1. Wi-Fi and wireless fundamentals

- **Wi-Fi:** Wireless LAN technology based on IEEE 802.11.
- **SSID:** Name identifying a Wi-Fi network.
- **Access Point:** Provides wireless access to a LAN.
- **Roaming:** Moving between access points/cells while maintaining network service.
- **WPA2/WPA3:** Wi-Fi security protocols.

Wi-Fi standards

Standard	Frequency / key point
802.11a	5 GHz; older standard
802.11b	2.4 GHz; older, lower-speed standard
802.11g	2.4 GHz; up to 54 Mbps
802.11n	2.4/5 GHz; MIMO
802.11ac	5 GHz; wider channels and higher throughput
802.11ax	2.4/5/6 GHz depending on implementation; Wi-Fi 6/6E, OFDMA

2. Bluetooth

- Short-range wireless communication.
- **Piconet:** Small Bluetooth network organized around a central device.
- **Scatternet:** Multiple interconnected piconets.
- **Frequency hopping:** Changes frequencies to reduce interference.

3. 5G network architecture

- **RAN (Radio Access Network):** Connects user devices through radio access.
- **Core network:** Handles connectivity, mobility and network services.

- **Network slicing:** Creates logically separated network instances for different requirements.
- 5G targets higher capacity, lower latency and massive device connectivity compared with earlier generations.

4. IoT communication

- **IoT:** Physical devices connected to networks to collect/exchange data.
- **MQTT:** Lightweight broker-based publish-subscribe protocol, commonly over TCP.
- **CoAP:** Lightweight REST-like request-response protocol for constrained devices, commonly over UDP.

MQTT	CoAP
Publish-subscribe	Request-response
Broker-based	Client-server
Commonly TCP	Commonly UDP
Lightweight telemetry	Constrained-device web transfer

5. Cloud networking and virtualization

- **IaaS:** Virtual machines, storage and network infrastructure.
- **PaaS:** Platform for building/deploying applications.
- **SaaS:** Ready-to-use software over a network.
- **CDN:** Delivers cached content from distributed locations closer to users.
- **Load balancing:** Distributes traffic across servers.
- **Virtual machine (VM):** Software-based computer with its own OS.
- **Hypervisor:** Creates/manages virtual machines.
- **Container:** Isolated application environment sharing the host OS kernel.
- **Container networking:** Connects containers to each other, hosts and external networks.

CHAPTER 5 — NETWORK SECURITY AND EMERGING TRENDS

New-syllabus chapter. The supplied PYQ analysis reports no previous-paper coverage for this chapter; prioritize the explicitly listed syllabus concepts.

1. Security fundamentals

- **Confidentiality:** Only authorized parties can access information.
- **Integrity:** Information is protected against unauthorized modification.
- **Availability:** Systems/services remain accessible when needed.
- **Authentication:** Verifies identity.
- **Authorization:** Determines permitted actions.
- **Non-repudiation:** Evidence supports the origin/action of a communication.

2. Cryptography

- **Encryption:** Converts plaintext to ciphertext.
- **Decryption:** Recovers plaintext.
- **Hashing:** Produces a fixed-length digest; it is not encryption.
- **Digital signature:** Supports authenticity, integrity and non-repudiation.

Symmetric encryption	Asymmetric encryption
Same secret key for encryption/decryption	Public/private key pair
Generally faster	Generally more computationally expensive
Example: AES	Example: RSA

3. Firewalls

- **Firewall:** Filters network traffic using security rules.
- **Packet-filtering firewall:** Checks packet headers and rules.
- **Stateful inspection firewall:** Tracks connection state.

- **Application-level gateway/proxy:** Inspects application-layer traffic.

Purpose: Allow authorized traffic and block traffic that violates policy.

4. VPN and secure communication

- **VPN:** Creates a protected logical connection over an untrusted network.
- **IPsec:** Protects IP traffic at the network layer.
- **TLS VPN:** Uses TLS protection for remote access.
- **Remote-access VPN:** Connects an individual user.
- **Site-to-site VPN:** Connects entire networks.
- **TLS:** Secures data in transit.
- **HTTPS = HTTP over TLS.**

Simplified TLS handshake

1. Client initiates handshake.
2. Server presents certificate.
3. Client validates certificate identity and trust.
4. Client/server establish shared session keys.
5. Encrypted communication begins.

5. IDS and IPS

- **IDS (Intrusion Detection System):** Detects suspicious activity and raises alerts.
- **IPS (Intrusion Prevention System):** Detects and can block malicious activity.
- **Signature-based detection:** Matches known attack patterns.
- **Anomaly-based detection:** Flags deviations from expected behavior.

6. Zero Trust

- Principle: **Never trust automatically; always verify.**

- Verify identity and device context.
- Apply least-privilege access.
- Continuously monitor access.
- Use micro-segmentation to limit lateral movement.

7. Common cybersecurity attacks

Attack	Meaning
DoS	Attempts to make a service unavailable
DDoS	Distributed traffic overwhelms a service
Man-in-the-Middle	Intercepts or may alter communication
Phishing	Deceptive messages trick users
Spoofing	Falsifies identity/address information
Sniffing	Captures network traffic
Botnet	Compromised devices controlled collectively
Malware	Software designed for harmful/unauthorized activity

8. SDN and AI-driven monitoring

- **SDN (Software-Defined Networking):** Separates control plane from data plane.
- **Control plane:** Decides how traffic should be forwarded.
- **Data plane:** Forwards packets.
- **SDN controller:** Logically central control of supported network devices.
- **OpenFlow:** Protocol that can communicate forwarding rules between controller and supported switches.
- **AI-driven monitoring:** Uses traffic analytics/anomaly detection to identify suspicious behavior.

FINAL QUICK REVISION

Highest-priority topics from the supplied PYQ analysis

1. CRC and codeword generation.
2. Distance Vector Routing and routing-table updates.
3. HTTP persistent/non-persistent connections and message format.
4. DNS hierarchy and resolution.
5. OSI vs TCP/IP model.
6. Socket programming system calls.
7. Go-Back-N vs Selective Repeat.
8. IP subnetting, CIDR and VLSM.
9. Link State Routing and Dijkstra.
10. CSMA variants, SMTP/POP3/IMAP, ALOHA, TCP header and congestion control.

Essential formulas

- Transmission delay = L/R.
- Propagation delay = D/S.
- Bit rate = Baud rate × bits per symbol.
- IPv4 subnet addresses = 2^(32 − prefix).
- Conventional usable hosts = 2^(host bits) − 2.
- GBN sender window max = 2^m − 1.
- SR window max = 2^(m−1).
- Pure ALOHA maximum efficiency ≈ 18.4%.
- Slotted ALOHA maximum efficiency ≈ 36.8%.

Final viva method

For each topic, be ready with:

1. **Definition** in one sentence.

2. **Working / key steps**.
3. **One comparison or advantage**.
4. **One example or diagram**, if relevant.
5. **Formula and a worked problem** for numerical topics.